

Case study Excel - 1

Aimee at Adventure Works has asked you to help her prepare an Excel file that she will present at a management team financial review meeting. The worksheet tracks the items sold by one of Adventure Work's divisions. The file needs to contain results such as costs, revenue, and profit. Your task is to create the formulas and calculations for this sheet.

Step 1: Download the file

Download and open the Microsoft Excel workbook *Revenue figures.xlsx*. The workbook contains only one worksheet called *Jan-Dec*.

Step 2: Create the calculations

1. Create a calculation in cell **G4** to calculate the Purchase Cost.

Tip: This would be the number of stock items purchased multiplied by the wholesale cost.

2. Create a calculation in cell **H4** that works out the total shipping costs for the number of items ordered. The shipping cost per item amount data is in cell **P1**.

Tip: Don't forget that you will be copying this formula down the column using Autofill. So, the **P1** reference needs to stay constant. (Make a note of how this formula is written.)

3. Create a calculation in **I4** that adds together the purchase and total shipping costs.
4. Create a calculation in **J4** that works out the retail price for the individual item in row 4. The retail price must be set high enough to cover the following:
 - a. The wholesale cost of each item.
 - b. The shipping cost for the individual item.
 - c. Plus a 50% markup.

Tip: Your formula needs to add the two costs together and then calculate a 50% increase. You'll need to use parentheses to ensure the calculation works correctly. Also, include dollar signs on the **P1** cell reference.

5. Create a calculation in **L4** that determines the amount of revenue the company earned by selling the number of items listed in **K4** by the retail price you have just worked out in **J4**.
6. Create a calculation in **M4** that subtracts the **Total Cost** figure from the **Revenue** figure to generate the **Profit** figure.
7. Use the Autofill shortcut on cell **G4** to copy the formula down through the column. Then use the double-click shortcut to repeat this action for the formulas in **H4, I4, J4, L4, and M4**.

Tip: When you double-click the shortcut for Autofill, Excel can use the blocks of completed cells to the left as a reference. As you are double-clicking to copy each formula, check which block of data Excel is using to decide where to stop the copy operation.

8. Cell **P3** must display the overall profit margin. A profit margin figure indicates what percentage of the sales turned into profit. The formula to calculate this is similar to the percentage difference formula already demonstrated. In this situation the formula will be:
=(Total Revenue – Total Costs)/Total Revenue

In cell **P3**, create this formula using the **Total Costs** total in cell **I201** and the **Revenue** total in cell **L201** to work out the profit margin